HEFT Variant Selection and Interference-Aware Routing for DAG Scheduling on Wireless Edge Networks

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May 4, 2026

1 Methodology

1.1 Simulator and Interference Model

All experiments use **ncsim**, a discrete-event simulator for DAG workflow scheduling on wireless edge networks [1]. ncsim models 802.11ax MAC-layer contention via the `csma_bianchi` interference model, which computes per-link throughput degradation as the product of an SINR factor (capturing hidden-terminal attenuation) and a Bianchi contention factor (capturing MAC-layer collisions from concurrent transmitters in the conflict graph). The effective per-flow bandwidth on a link with k active neighbours in the conflict graph is:

$$B_{\text{eff}} = \frac{B_{\text{PHY}} \cdot f_{\text{SINR}} \cdot f_{\text{Bianchi}}(k)}{n_{\text{flows}}}$$

where n_{flows} is the number of concurrent flows on the link and $f_{\text{Bianchi}}(k)$ is the normalised throughput from the Bianchi model for k contending stations.

RF parameters are fixed throughout: $P_{\text{tx}} = 20\,\text{dBm}$, $f = 5\,\text{GHz}$, path-loss exponent $n = 3.0$, channel width 20 MHz, Wi-Fi standard 802.11ax. Adjacent links on a 40 m grid have a PHY rate of approximately 8.6 MB/s.

1.2 Network Topologies

We evaluate on two classes of topology:

Grid networks. Two grid topologies with *checkerboard diagonal* links for additional routing diversity:

- 4×4 grid: 16 nodes, 68 bidirectional links, grid spacing 40 m.
- 7×7 grid: 49 nodes, 240 bidirectional links, grid spacing 40 m.

Node compute capacities are heterogeneous in the range 80–300 cu/s.

Random geometric graphs (RGG). Fifty nodes are placed uniformly at random in an $L \times L$ square; bidirectional links are created between all pairs within communication range $R = 80\,\mathrm{m}$. Connectivity is guaranteed by bridging any disconnected component with its nearest neighbour. We sweep $L \in \{150, 200, 250, 300, 350, 400, 500\}$ m, yielding average node degrees ranging from 24.0 (L150, dense) to 3.7 (L500, sparse). The topology seed is fixed at 42 per density level; 30 independent workload seeds are used per configuration.

1.3 DAG Workloads

Each DAG uses heterogeneous task compute costs (150–1000 cu) and edge data sizes (2–30 MB), giving a compute-to-communication ratio range of approximately 0.1–100.

DAG	Structure	Tasks	Edges
Small	Fork-join (1 source → 6 parallel → 1 sink)	8	12
Large (4×4)	5-stage pipeline with cross-connections	30	48
Large (7×7)	6-stage pipeline with cross-connections	60	102

Table 1: DAG workloads used in experiments.

1.4 Scheduler Variants

Both variants use HEFT (Heterogeneous Earliest Finish Time) [2] but differ in the pairwise bandwidth matrix passed to the scheduler:

Variant	Adjacent BW	Non-adjacent BW
HEFT-1	Direct-link PHY rate	0.001 MB/s (heavy penalty)
HEFT-2	Widest-path PHY rate	Widest-path PHY rate

Table 2: Pairwise bandwidth matrices. HEFT-1's heavy non-adjacent penalty strongly biases placement toward co-location or direct-neighbour assignment.

The heavy penalty in HEFT-1 causes the scheduler to prefer assigning communicating task pairs to the same node (co-location) or to direct neighbours, minimising the number of wireless hops and hence interference exposure. HEFT-2 uses realistic widest-path estimates for all pairs and therefore distributes tasks across the topology.

1.5 Routing Schemes

We evaluate nine routing schemes spanning three categories:

Label	Name	Decision criterion
W	Widest-Path	Maximise bottleneck bandwidth
S	Shortest-Path	Minimise total latency ($\approx$ min hops)
SH	Shortest-Hop	Minimise hop count (tie-break: $\sum 1/b_\ell$)
GS	Greedy-Start	Static greedy; flows sorted by estimated start time
GC	Greedy-Criticality	Static greedy; flows sorted by HEFT upward rank
GB	Greedy-Bytes	Static greedy; flows sorted by data size (desc.)
GO	Greedy-Overlap	Static greedy; flows sorted by link overlap degree
GSD	Dynamic	Routes computed at transfer start using live link state
GSD-D	Dynamic + Deferral	GSD with deferral when congestion exceeds threshold

Table 3: Routing schemes evaluated.

1.6 Experimental Design

RQ1 — HEFT-1 vs. HEFT-2. To answer RQ1 we conduct a full factorial experiment on grid topologies: 2 schedulers $\times$ 9 routing $\times$ 4 grid experiments $\times$ 30 seeds = 2,160 ncsim runs. We additionally compare SAGA's *predicted* makespan (computed analytically from each scheduler's static BW model, with no simulation) against the ncsim *measured* makespan. This exposes the direction and magnitude of the prediction error.

For random networks we run 2 schedulers $\times$ 7 densities $\times$ 2 DAG sizes $\times$ 30 seeds = 840 runs per routing scheme. Results are reported for shortest-hop (SH) routing as a fixed, topology- agnostic baseline to isolate scheduler effects from routing effects.

RQ2 — Routing for HEFT-1. To answer RQ2 we fix the scheduler to HEFT-1 and compare all nine routing schemes across all grid experiments ($9 \times 4 \times 30$ = 1,080 runs) and across all random network configurations ($9 \times 7 \times 2 \times 30$ = 3,780 runs). Mean makespan and 95 % confidence intervals (CI) over 30 seeds are reported.

2 Results

2.1 RQ1: HEFT-1 vs. HEFT-2 — SAGA Prediction vs. ncsim Reality

2.1.1 SAGA predicts HEFT-2 wins by orders of magnitude

Table 4 shows SAGA's analytically predicted makespan for each scheduler on grid experiments. Because HEFT-2 estimates widest-path PHY rates for all pairs, it predicts cheap transfers and produces short makespans. HEFT-1 penalises non-adjacent pairs at 0.001 MB/s, so SAGA sees a massive transfer bottleneck wherever tasks are not co-located, inflating its predicted makespan by 168–727$\times$ relative to HEFT-2.

Experiment	HEFT-1 (SAGA, s)	HEFT-2 (SAGA, s)	Ratio H1/H2
4×4, small	11.0	**10.6**	1.04$\times$
4×4, large	4,021.9	**23.9**	168$\times$
7×7, small	10.9	**10.2**	1.07$\times$
7×7, large	18,039.0	**24.8**	727$\times$

Table 4: SAGA-predicted makespan. **Bold green**: lower (predicted winner). On large DAGs, SAGA predicts HEFT-2 is 168–727$\times$ faster than HEFT-1.

2.1.2 ncsim reverses the ranking: HEFT-1 wins by 3–5×

Table 5 shows ncsim-measured makespan with shortest-hop (SH) routing. The ranking inverts completely: HEFT-1 is 3.7–5.1× faster than HEFT-2 across all grid experiments. With best-routing per scheduler (Table 6), the gap is 2.9–5.1×.

Experiment	HEFT-1 (SH, s)	HEFT-2 (SH, s)	Ratio H2/H1
4×4, small	**18.3**	93.9	5.1×
4×4, large	**292.4**	1,071.7	3.7×
7×7, small	**20.3**	93.8	4.6×
7×7, large	**614.2**	2,492.1	4.1×

Table 5: ncsim-measured makespan with SH routing (mean over 30 seeds). **Bold green**: lower (actual winner). HEFT-1 wins in every experiment, reversing SAGA's prediction.

Exp.	H1 best rt.	H1 ms (s)	H2 best rt.	H2 ms (s)	Ratio H2/H1
4×4 S	GSD-D	**18.0**	GSD-D	91.1	5.1×
4×4 L	GS	**292.1**	GSD	855.9	2.9×
7×7 S	GSD-D	**19.8**	SH	93.8	4.7×
7×7 L	GO	**532.5**	S	2,461.9	4.6×

Table 6: Best-routing ncsim results per scheduler. HEFT-1 wins in all four experiments. The best routing scheme for each scheduler differs, but the scheduler ranking is stable.

2.1.3 Mechanism: co-location eliminates wireless transfers

The inversion is explained by the extra metrics extracted from each SAGA schedule (Table 7). HEFT-1's penalty forces near-total co-location on small DAGs (67 % of edges) and partial co-location on large DAGs, reducing mean wireless hops to 0.33–1.20. With few or no cross-node transfers, there is no interference. HEFT-2's widest-path estimates encourage spreading tasks across the topology: mean hops are 2.67–4.22, generating dense concurrent transfers whose throughput collapses under CSMA/CA contention.

Experiment	Scheduler	Co-loc. fraction	Mean hops	Cross-node MB
4×4, small	HEFT-1	0.67	0.33	30
	HEFT-2	0.42	4.00	124
4×4, large	HEFT-1	0.40	0.71	—
	HEFT-2	0.42	2.67	—
7×7, small	HEFT-1	0.67	0.50	—
	HEFT-2	0.42	3.17	—
7×7, large	HEFT-1	0.32	1.20	—
	HEFT-2	0.38	4.22	—

Table 7: Co-location and hop metrics from SAGA schedule analysis. HEFT-1 routes substantially fewer hops, directly reducing interference exposure.

2.1.4 HEFT-1 dominates across all random network densities

Figure 1 shows ncsim makespan with SH routing on random geometric graphs for the large DAG (30 tasks). HEFT-1 is consistently lower than HEFT-2 across all seven density levels; the gap widens as the network becomes sparser (Table 8). At L150 (dense, avg. degree 24.0) HEFT-2 is 2.5× slower; at L500 (sparse, avg. degree 3.7) it is 5.7× slower. The widening gap at low density occurs because sparser networks force HEFT-2 to use longer multi-hop paths, compounding interference at each relay.

Density	Avg. deg.	HEFT-1 (s)	HEFT-2 (s)	Ratio H2/H1
L150	24.0	**148.6**	375.8	2.5×
L200	17.1	**289.3**	768.4	2.7×
L250	12.4	**325.6**	1,108.9	3.4×
L300	8.7	**328.4**	1,609.4	4.9×
L350	6.7	**337.2**	2,340.7	6.9×
L400	5.8	**473.2**	2,558.8	5.4×
L500	3.7	**684.5**	3,869.9	5.7×

Table 8: ncsim makespan with SH routing, large DAG, random networks. HEFT-1 wins at every density level; the advantage grows as the network becomes sparser and paths lengthen.

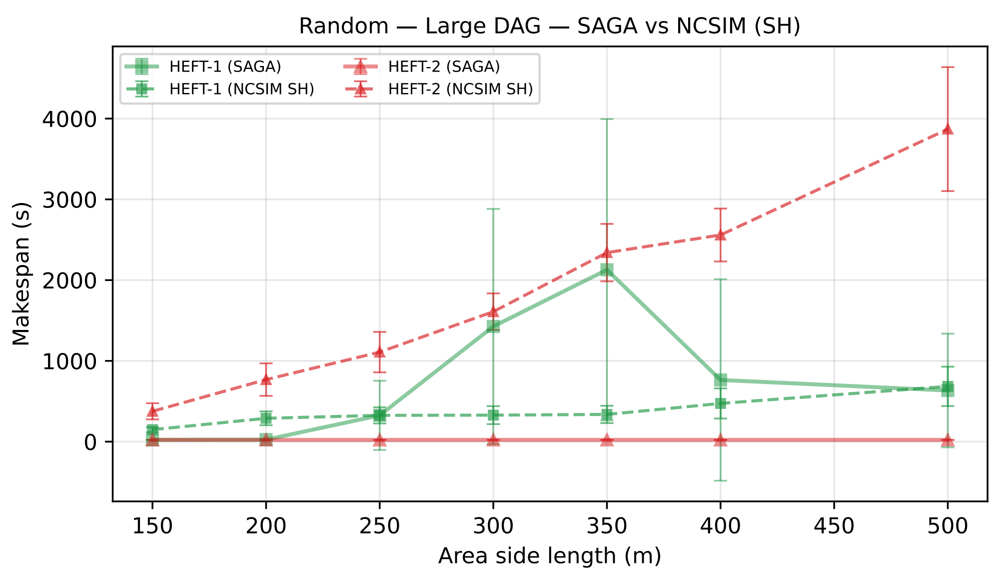

Figure 1: ncsim makespan (SH routing, solid) and SAGA predicted makespan (dashed) vs. network density for the large DAG. Green = HEFT-1, red = HEFT-2. HEFT-1 consistently achieves lower measured makespan while SAGA predicts the opposite.

2.2 RQ2: Which Routing Scheme Works Best with HEFT-1?

2.2.1 Routing matters less for HEFT-1 than for HEFT-2

Because HEFT-1 co-locates the majority of task pairs, most edges generate no wireless transfer at all. On small DAGs (8 tasks, fork-join), all routing schemes produce nearly identical makespans under

HEFT-1 (Table 9): the variation across all nine schemes is under 2 s on both grid sizes, confirming that routing is irrelevant when transfers are rare. On large DAGs, routing begins to matter because a larger DAG forces more cross-node transfers even under HEFT-1's aggressive co-location bias.

2.2.2 Widest-path is the worst choice, even for HEFT-1

Widest-path routing (W) is the worst-performing scheme in every HEFT-1 experiment on large DAGs (Table 9). On the 7×7 large experiment, W achieves 1,346 s vs. GO's 532 s — a 2.5× penalty. Even the small number of cross-node transfers that HEFT-1 generates are routed through long, bandwidth-greedy multi-hop paths that accumulate severe interference. Minimising hop count is therefore more important than maximising bottleneck bandwidth under CSMA/CA interference.

Routing	Small DAG		Large DAG	
	4×4 (s)	7×7 (s)	4×4 (s)	7×7 (s)
W	18.3	24.2	497.8	1,346.3
S	18.3	20.3	292.4	618.5
SH	18.3	20.3	292.4	618.5
GS	18.3	20.3	**292.1**	619.0
GC	18.3	20.3	292.1	641.4
GB	18.3	20.3	292.1	600.6
GO	18.3	20.3	292.1	**532.5**
GSD	18.3	20.3	671.4	20.3
GSD-D	**18.0**	**19.8**	322.5	1,537.8

Table 9: HEFT-1 makespan by routing scheme on grid experiments (mean over 30 seeds). **Bold green**: best. Red: worst. On small DAGs all schemes are nearly equivalent. On large DAGs W is the clear worst; GS and GO win on 4×4 and 7×7 large respectively.

2.2.3 No single scheme universally dominates, but SH is a safe default

On grid networks, the best HEFT-1 routing scheme depends on the regime: GSD-D wins on small DAGs by deferring the few cross-node transfers until contention subsides; GS wins on the 4×4 large DAG; and GO wins on the 7×7 large DAG where the larger grid provides enough path diversity for overlap-based greedy ordering to pay off.

On random geometric graphs (Table 10), S and SH consistently achieve the lowest or near-lowest makespan for HEFT-1 across all seven density levels. SH wins or ties for best in 6 of 7 densities on the small DAG and ties for best in 5 of 7 on the large DAG. Shortest-hop routing avoids accumulating interference across relay nodes, which is the dominant cost even for the sparse cross-node transfers generated by HEFT-1.

Density	Best (small)	ms (s)	Best (large)	ms (s)
L150	S	21.6	S	—
L200	S	27.9	S	—
L250	S	23.5	S	—
L300	SH	24.4	SH	—
L350	SH	24.2	SH	—
L400	S	22.9	S	—
L500	S	20.9	GS	—

Table 10: Best routing scheme for HEFT-1 per density level on random networks (from random_network_results.pdf Table 3). S and SH dominate across all densities.

2.2.4 Summary: routing recommendations for HEFT-1

- **Small DAGs**: routing is irrelevant under HEFT-1 (all schemes within $\approx 2\,\mathrm{s}$ of each other). Use any scheme; GSD-D provides a marginal advantage.
- **Large DAGs, dense or regular topology**: use a greedy scheme (GS or GO) that accounts for concurrent transfer overlap.
- **Large DAGs, arbitrary or random topology**: use SH or S. Min-hop routing is a robust, topology-agnostic choice that avoids interference accumulation without requiring knowledge of concurrent flow state.
- **Avoid W**: widest-path routing is the worst choice in every large-DAG experiment regardless of topology, because bandwidth-greedy paths are long and accumulate interference at each relay hop.

Acknowledgements

This work was conducted at the Autonomous Networks Research Group (ANRG), University of Southern California.

References

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[2] H. Topcuoglu, S. Hariri, and M.-Y. Wu, "Performance-effective and low-complexity task scheduling for heterogeneous computing," *IEEE Trans. Parallel Distrib. Syst.*, vol. 13, no. 3, pp. 260–274, Mar. 2002.